

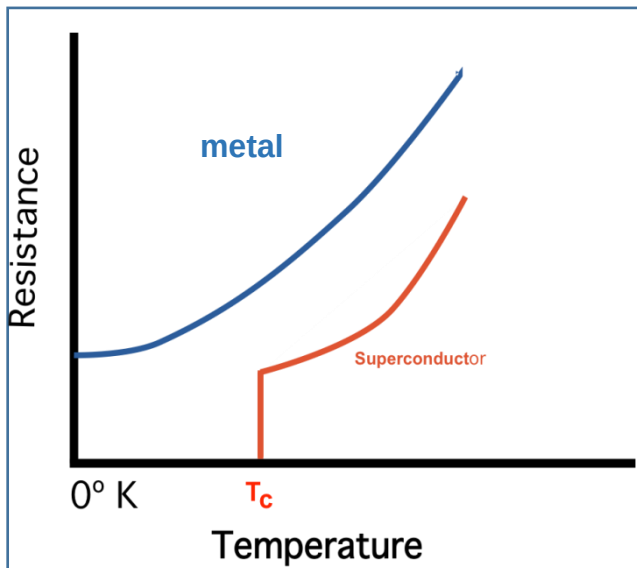
High temperature Superconducting materials, functionalities and devices at ICMAB

T Puig

**X. Obradors, X. Granados, A. Palau, J. Gutierrez, S. Ricart,
M Coll, J. Gazquez**

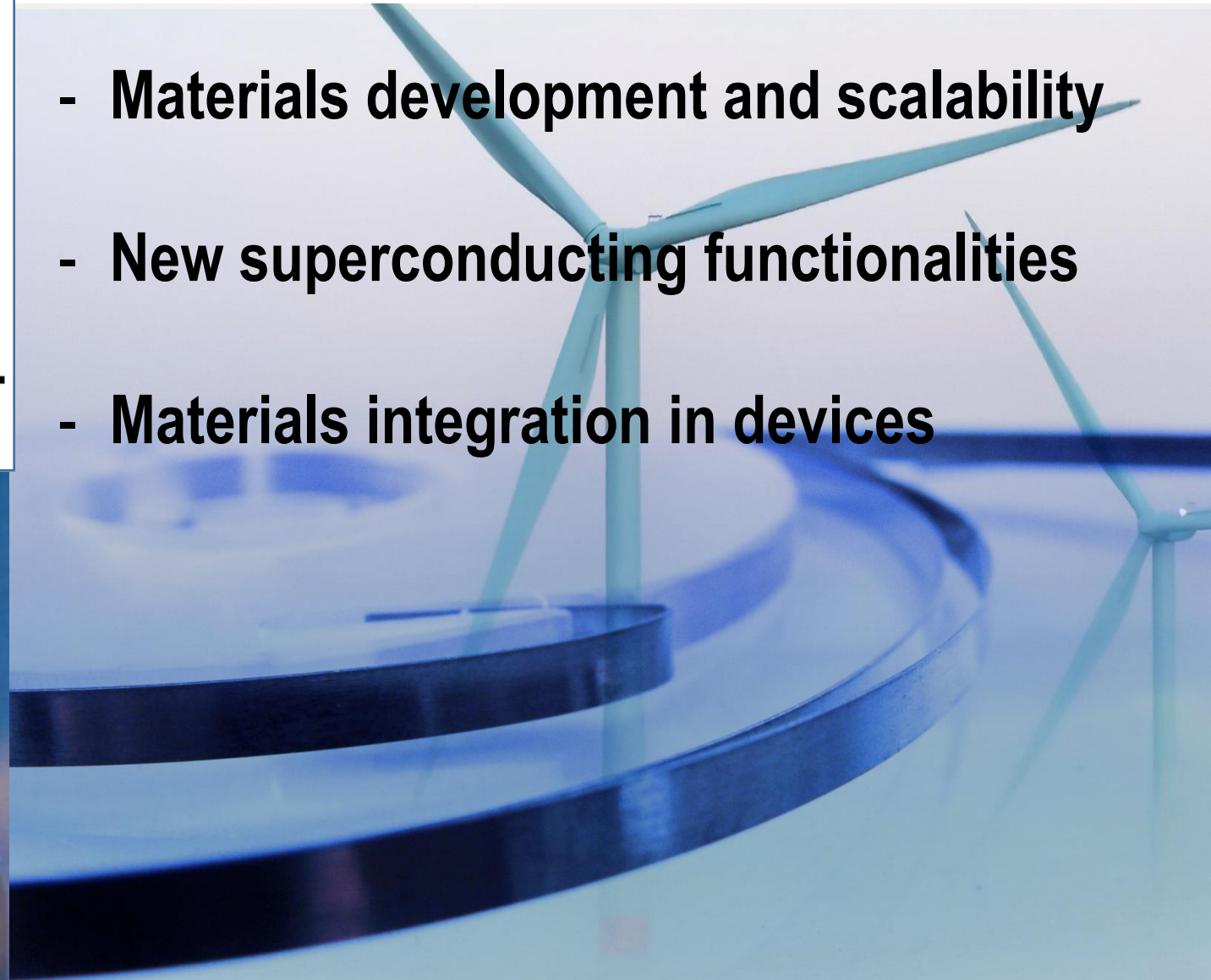
Superconducting Materials and large scale nanostructures Department
Institut de Ciència de Materials de Barcelona, CSIC, Bellaterra, Spain

RL2: Superconducting materials for power applications



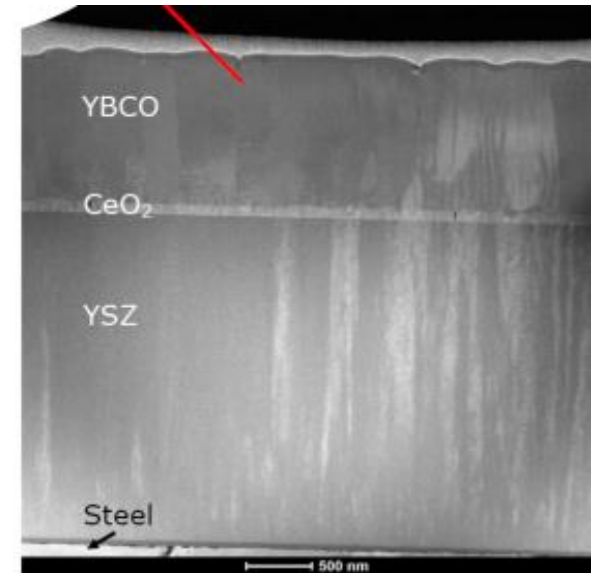
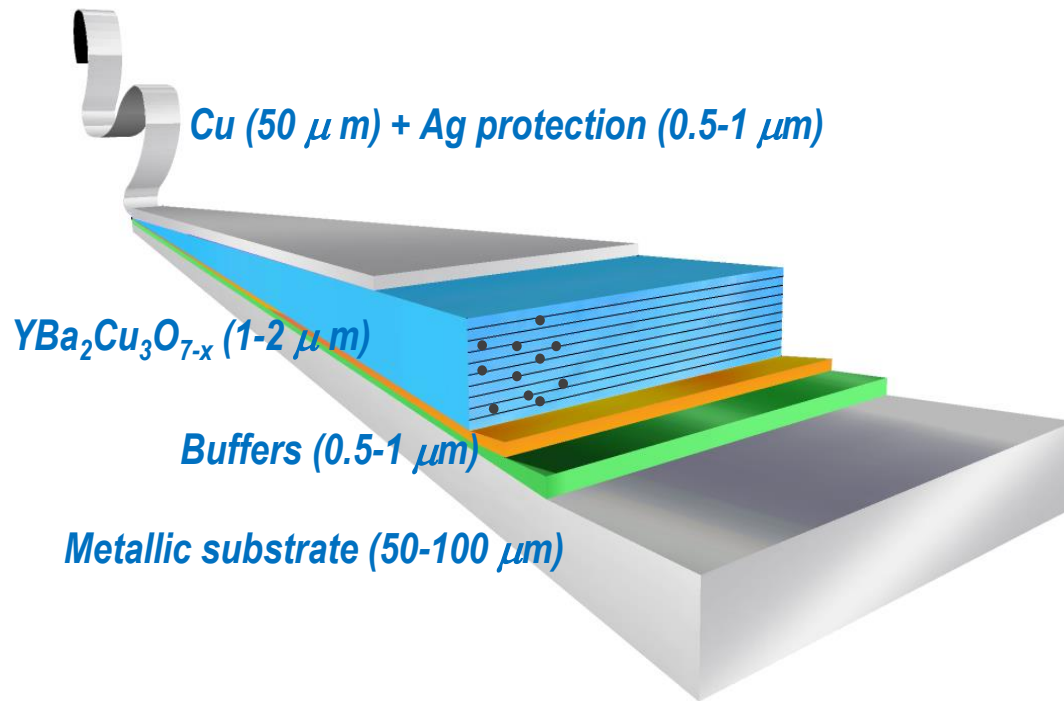
- Materials development and scalability
- New superconducting functionalities
- Materials integration in devices

1000 A
equivalents:
Cu and HTS



High Temperature Superconducting Coated Conductors

A revolution in materials science



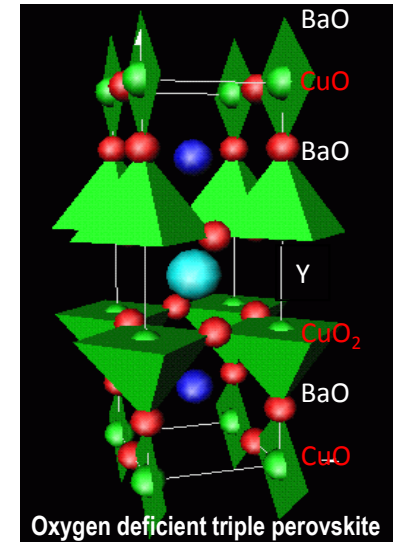
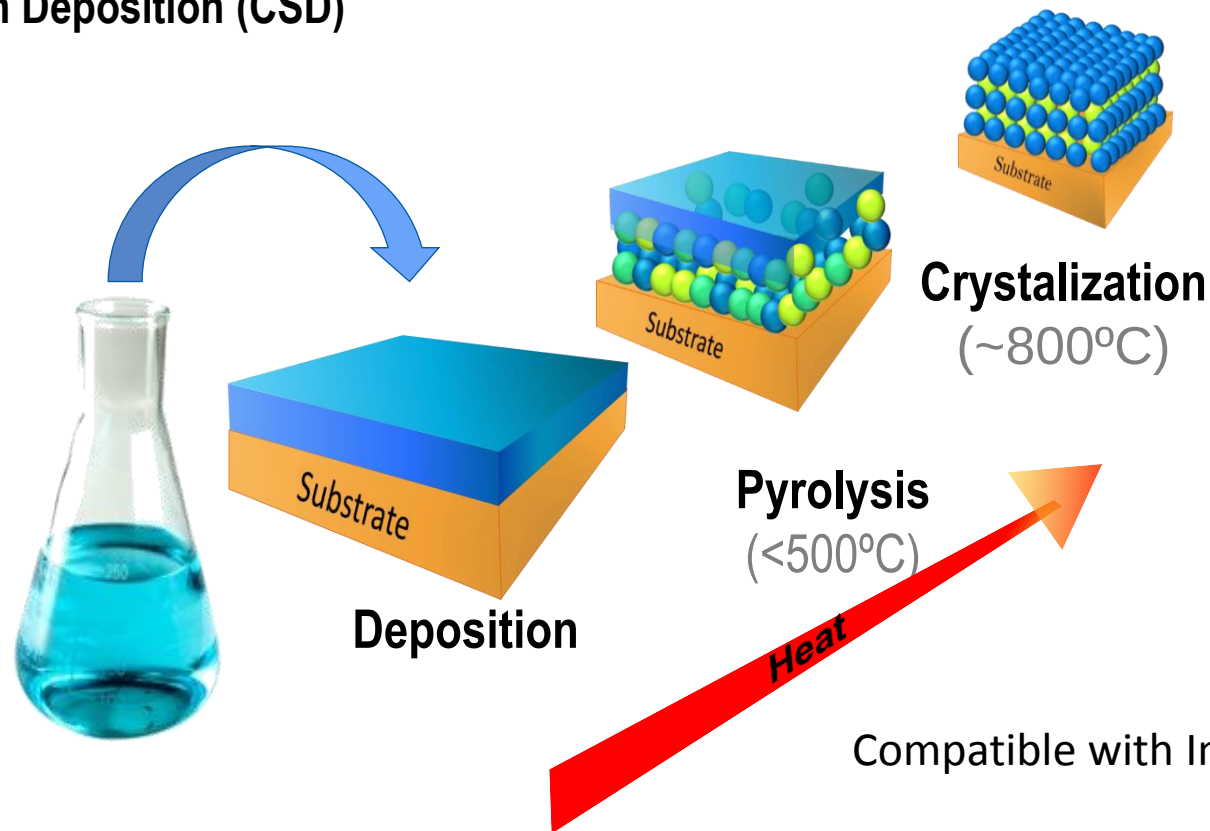
Km-length flexible epitaxial multilayer capable to carry 400 A/cm-w at 77K and 800 A/cm-w at 5 K & 30 T

Coated Conductors at ICMAB

(17 years of experience)

Low cost, scalable, high –throughput / high performance HTS-cuprate materials, film growth and materials physics understanding

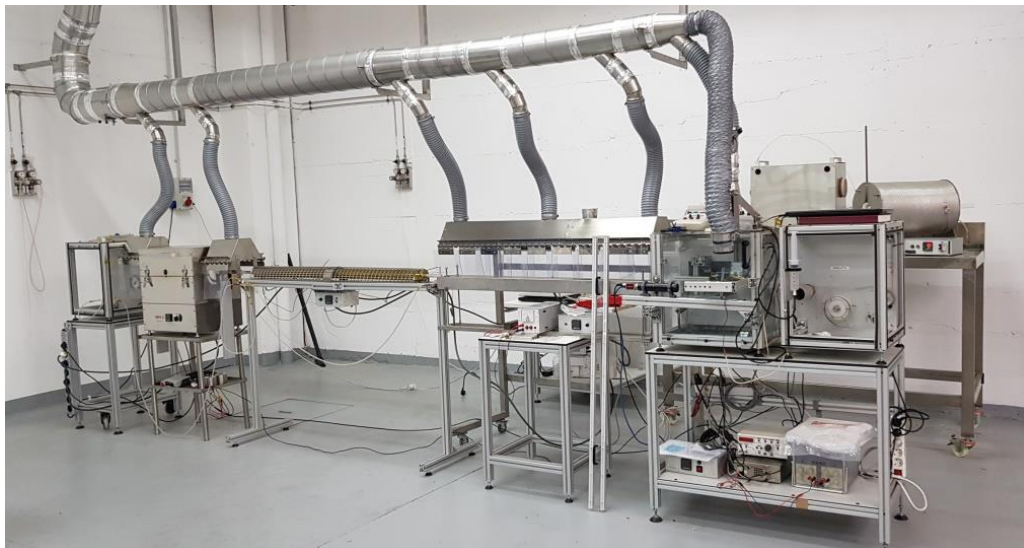
Chemical Solution Deposition (CSD)



Compatible with Ink Jet Printing

Spin-off created in 2010
5 employees
500m² in a industrial park

Oxide layers by CSD- ink jet printing
Coated Conductors for sustainable development

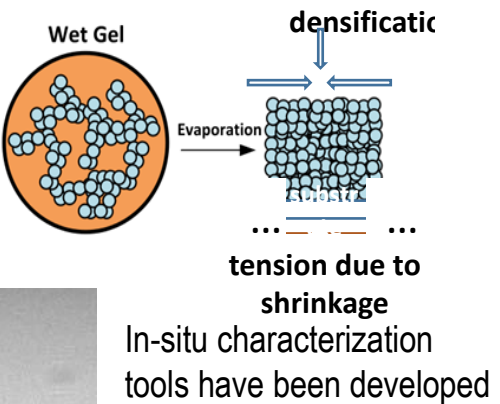
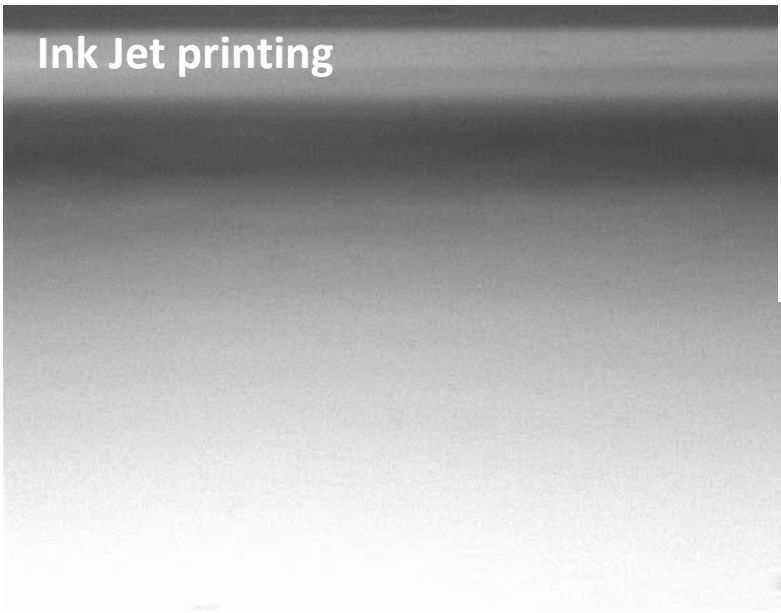


SUPERINKS: MINECO-RTC-Advanced Inks

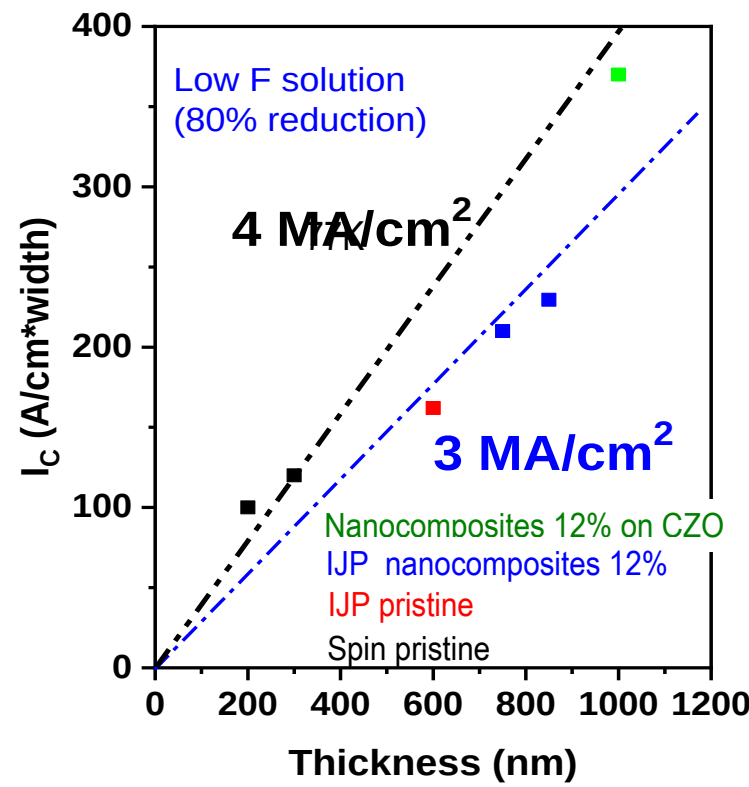
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Development of UV inks for coated conductors deposition

Towards growth of thick coated conductors: Scalability by innovative inks

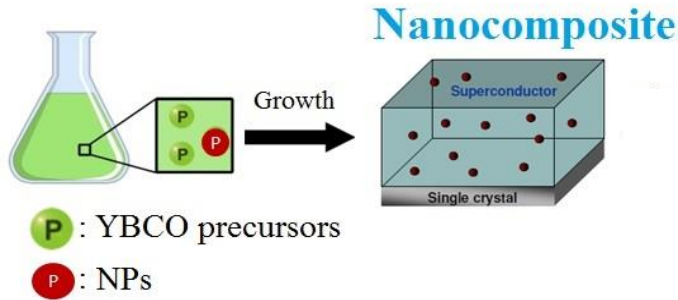


$I_c = J_c \times \text{thickness}$

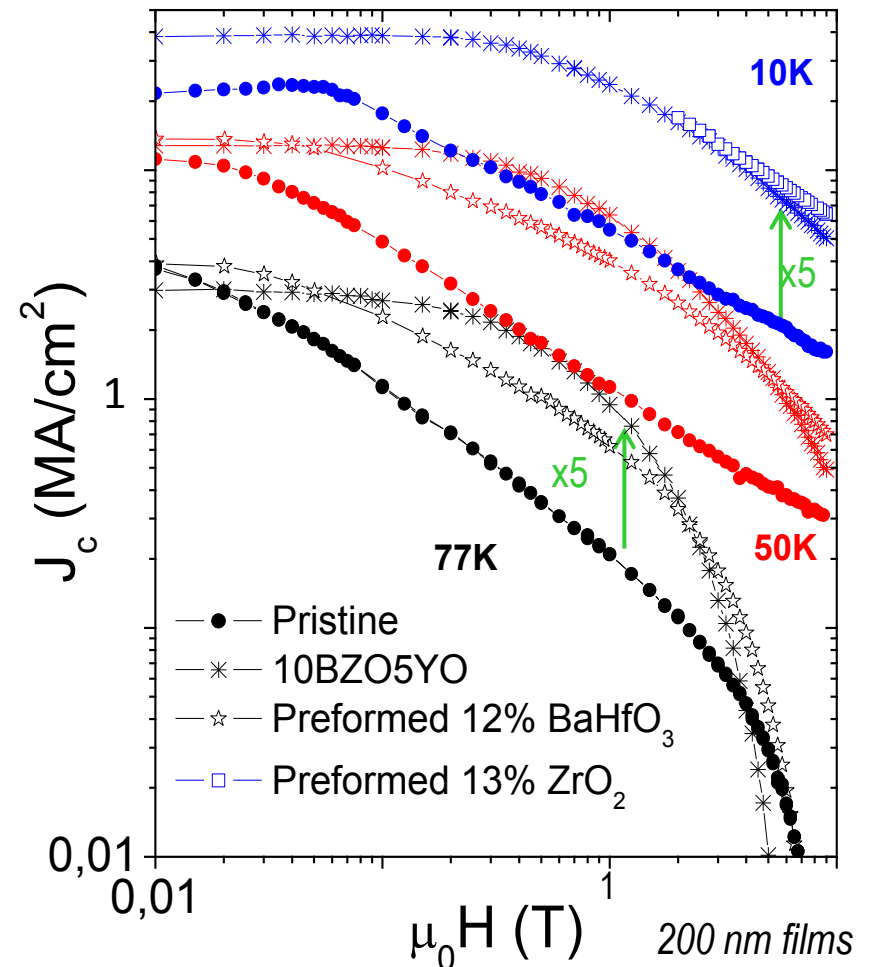
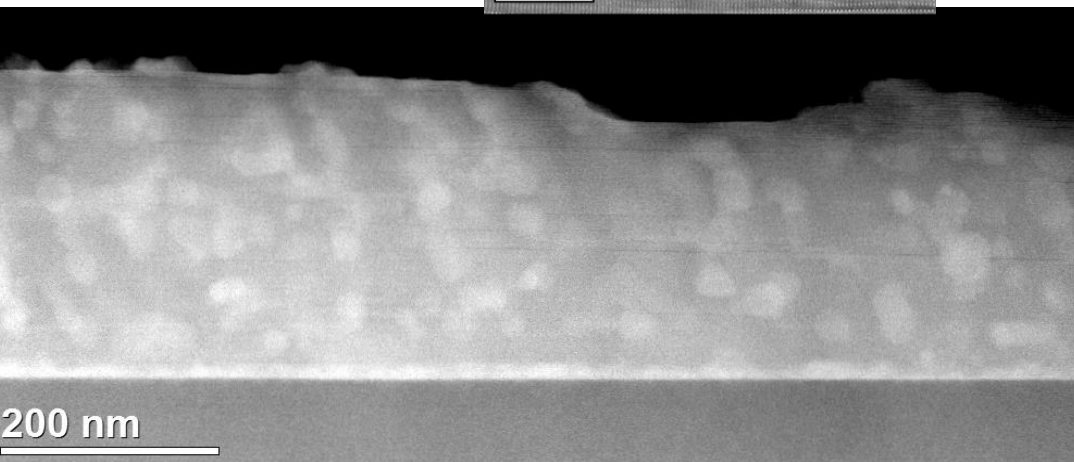
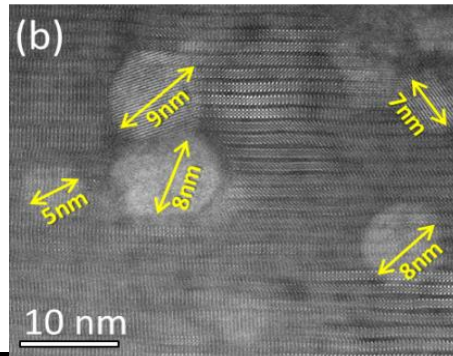
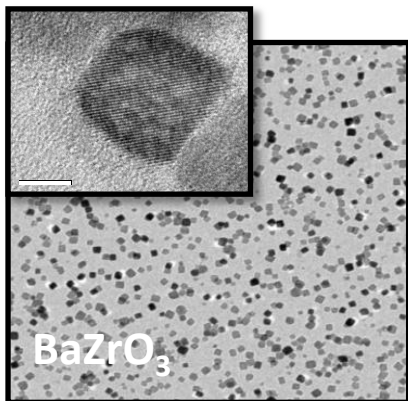


EUROTAPES: FP7 EU Integrated project

X. Obradors ICMAB coord. 21 partner (8 industrial)

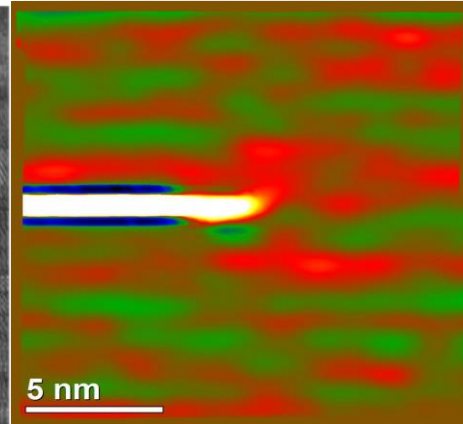
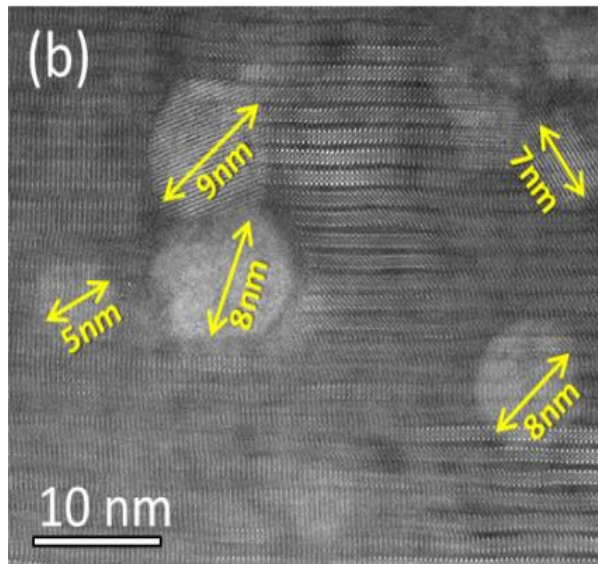


Integrating nanocomposite in coated conductors with cost effective processes

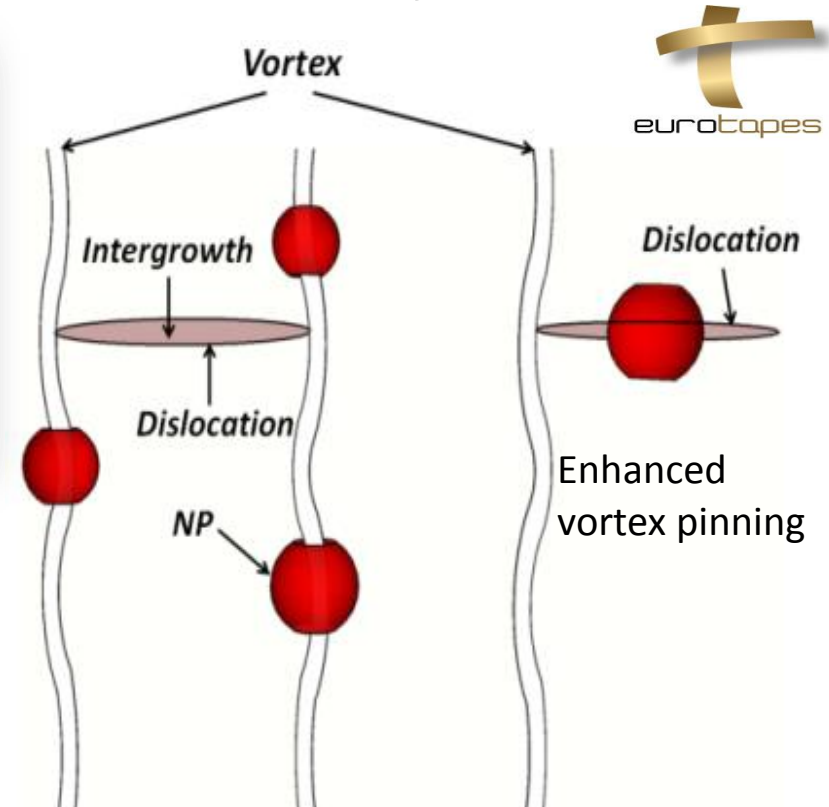


Nobel vortex pinning functionalities

Nanotechnology is key



Local strain effects generated at the partial dislocations



J. Gutierrez et al Nat Mat. (2007)

A. Llordés et al Nat Mat (2012)

Z. Li et al, to be submitted

Nanostrain and NPs (4-8 nm) Synergistic effect for enhanced vortex pinning

Control of the overdoped state should further increase performance

ULTRASUPERTAPE: ERC Advanced Grant H2020

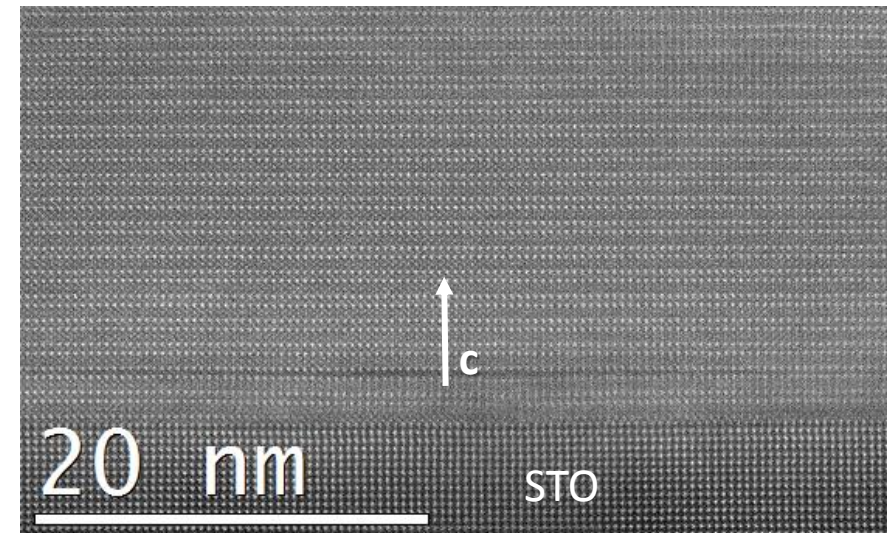
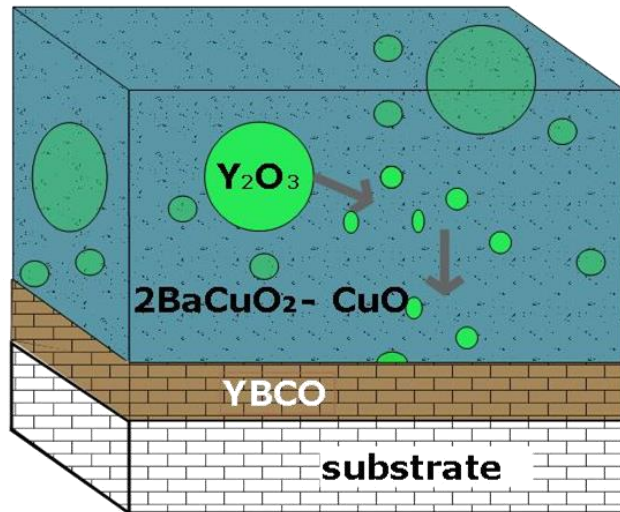
T. Puig

Ultrafast growth of ultrahigh performance superconducting tapes

- **Ultrafast CSD-Transient Liquid Assisted Growth ($\times 100$)**
- **Thick Strained Overdoped Nanocomposites ($\times 100$)**
- **Additive Manufacturing (*scalable*)**
- **New combinatorial chemistry approaches (fast screening)**



Transient Liquid Assisted Growth: TLAG



$$J_c(77K) = 4-5 \text{ MA/cm}^2$$

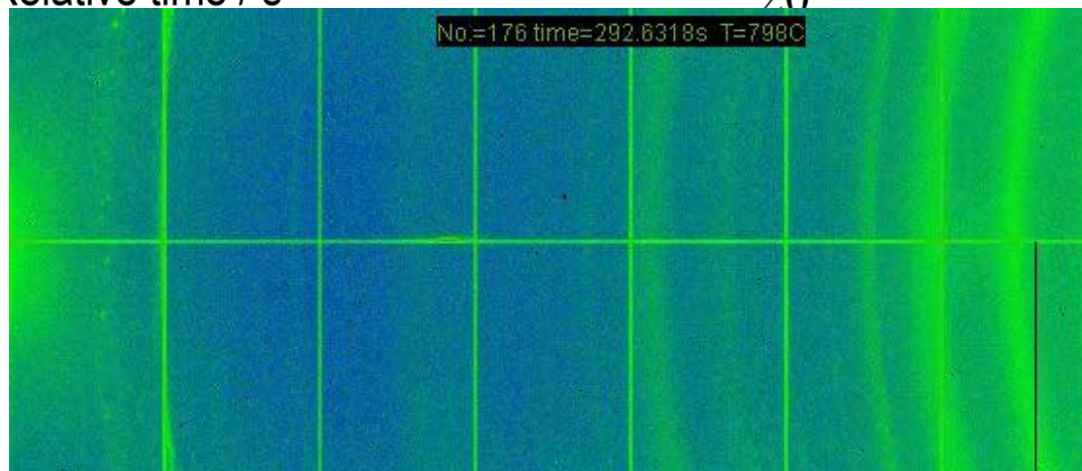
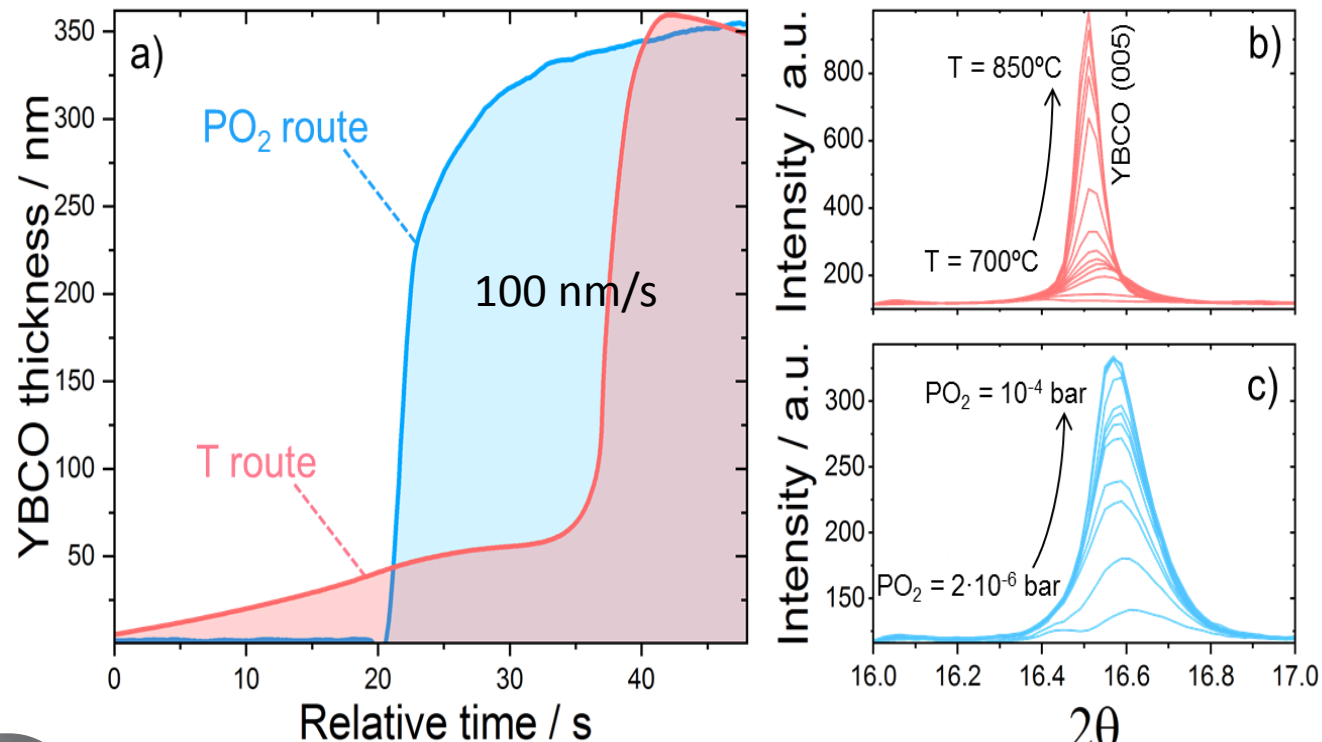
In-situ XRD
synchrotron exp.

Growth rate x100
standard thin film growth
processes

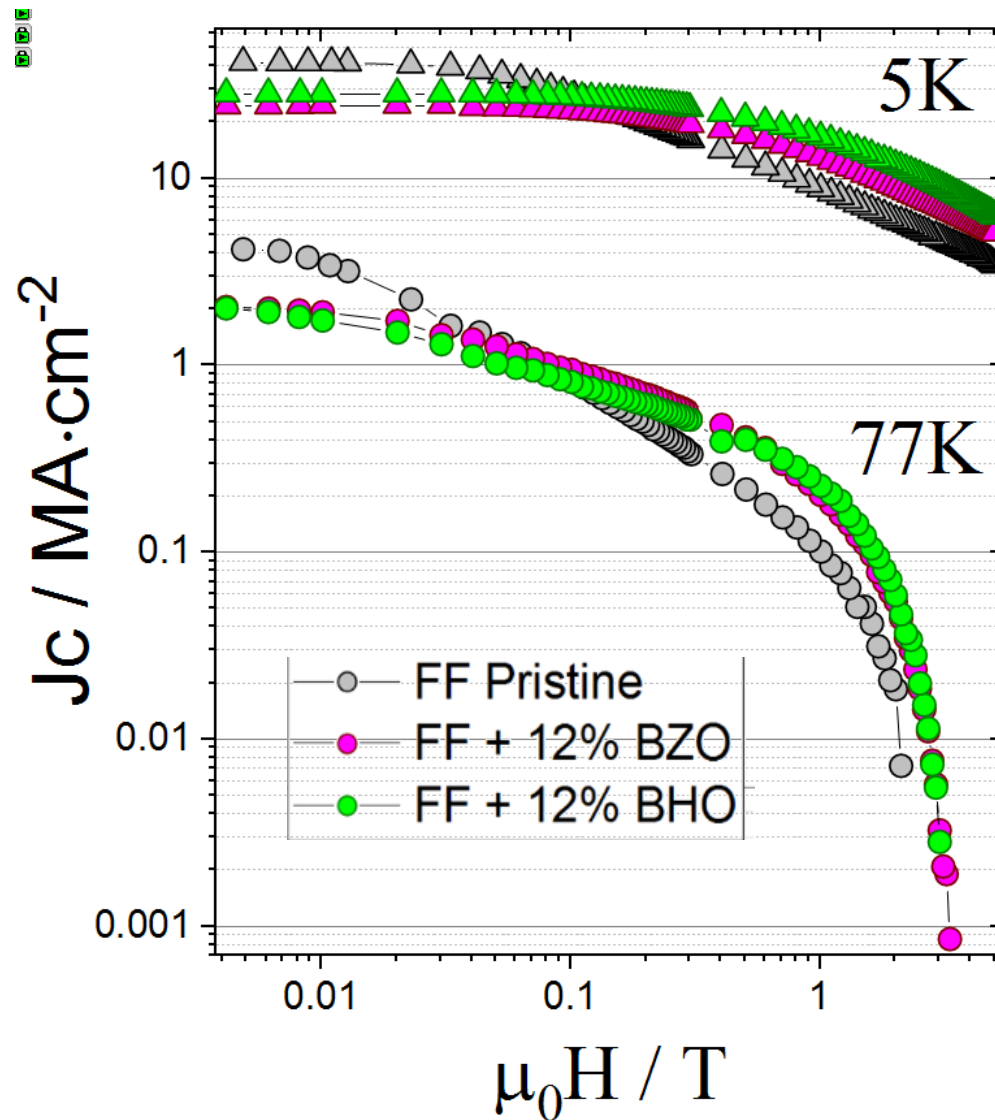
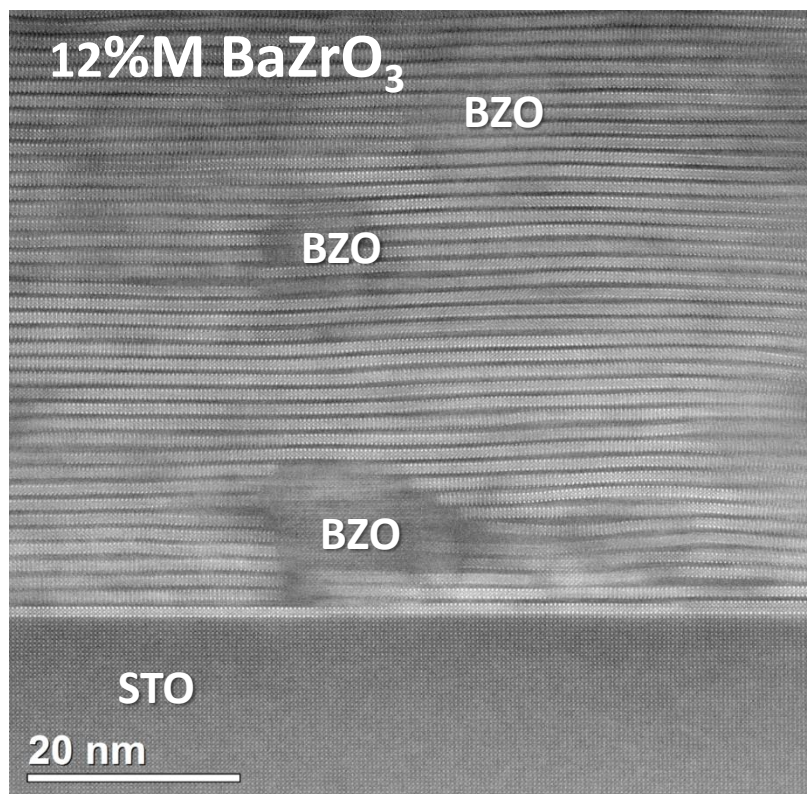
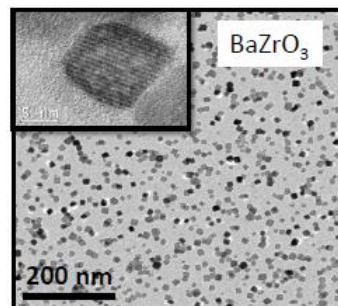
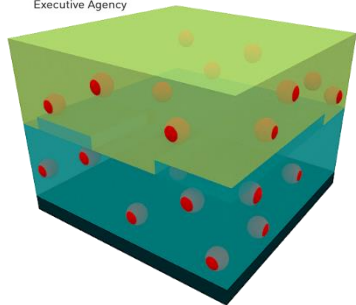
DiffAbs beamline
100 ms acquisition time



L. Soler et al, to be published



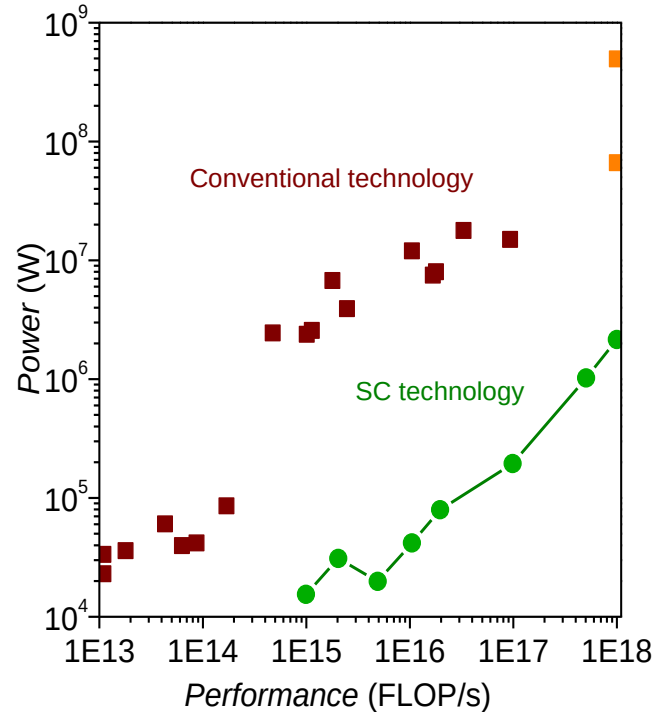
Nanocomposites by TLAG-CSD



New superconducting functionalities for green energy solid state electronic devices

Superconducting technology:

by virtue of their inherent energy-efficiency emerges as a very promising and reliable approach to **close the growing gap between required performance and sustainability**



www.top500.org, Holmes et al. IEEE Trans. Appl. Supercond. 23, 1701610 (2013)

Two energy efficient approaches to encode and manipulate information.



Nanoscale electric field-induced oxygen doping modulation:
Resistive switching phenomena

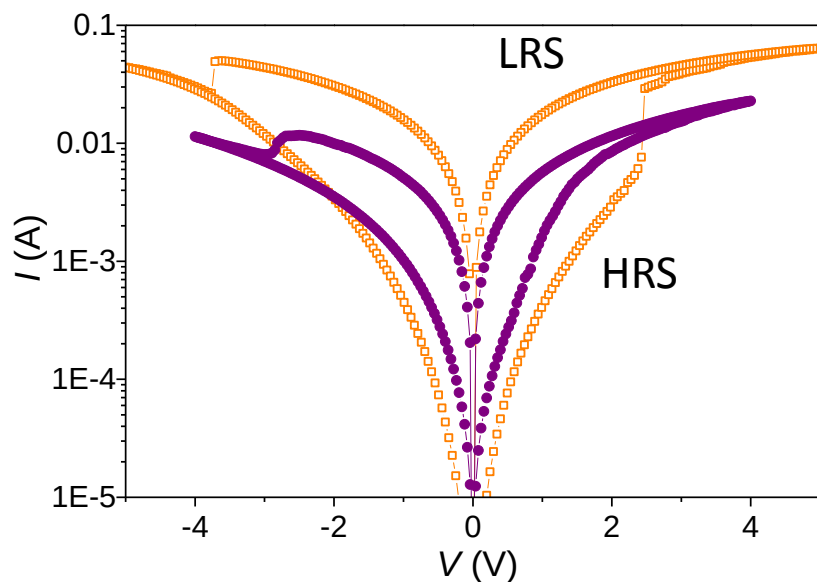
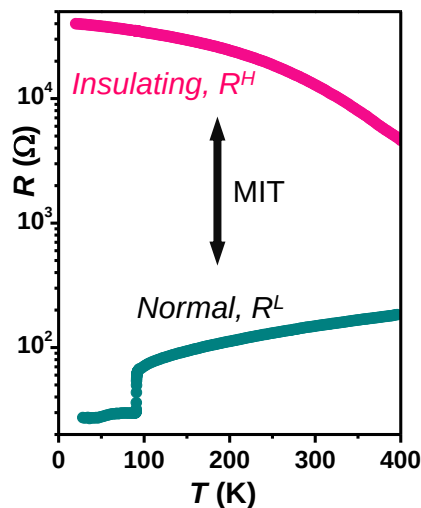
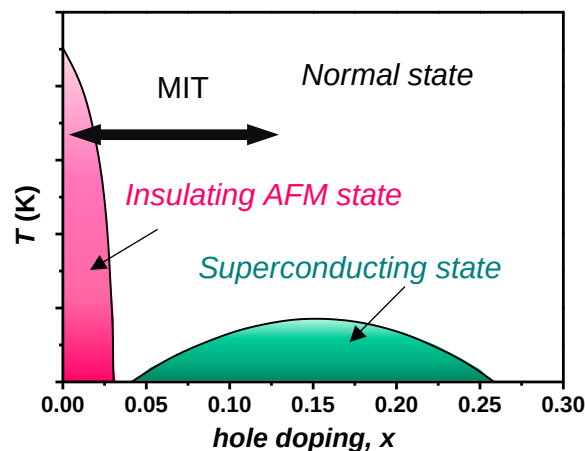


Magnetic spin modulation with supercurrents (trapped fields):
Hybrid SC / FM systems

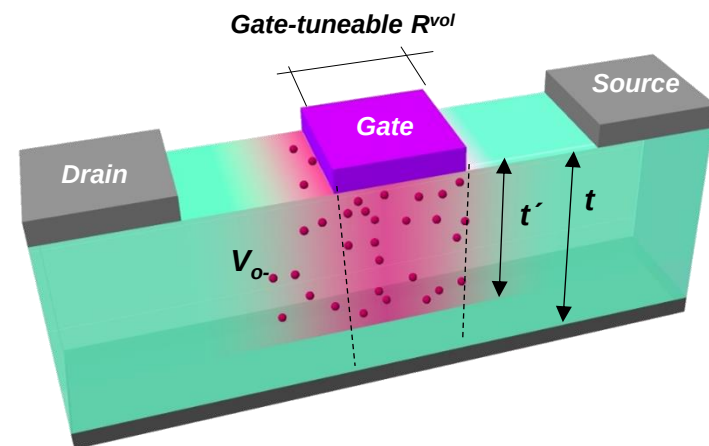
Resistive Switching in YBCO Films

A. Palau and N. Mestres FIP proj. (SO)

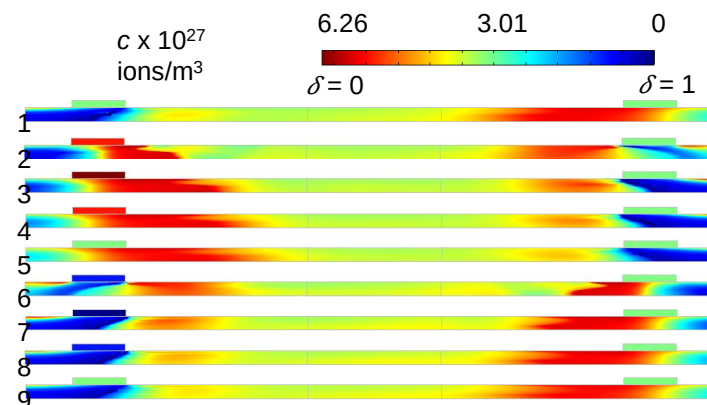
Reversible field induced Metallic / Insulating transition



Transistor like devices with a free-resistance channel



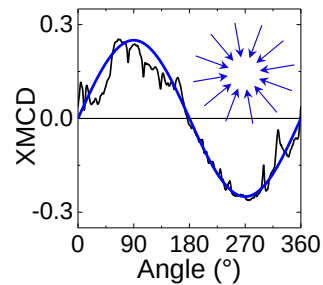
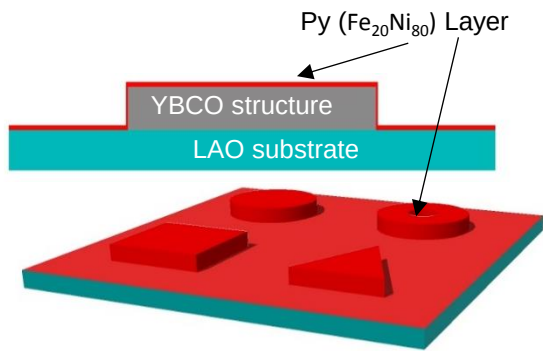
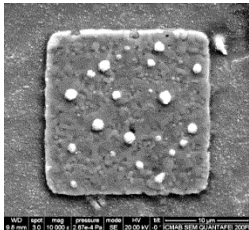
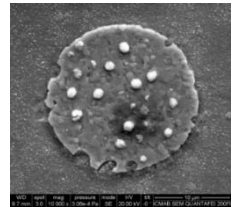
Homogenously and reversibly tune a superconducting channel on and off, by means of an electric field as the external control parameter.



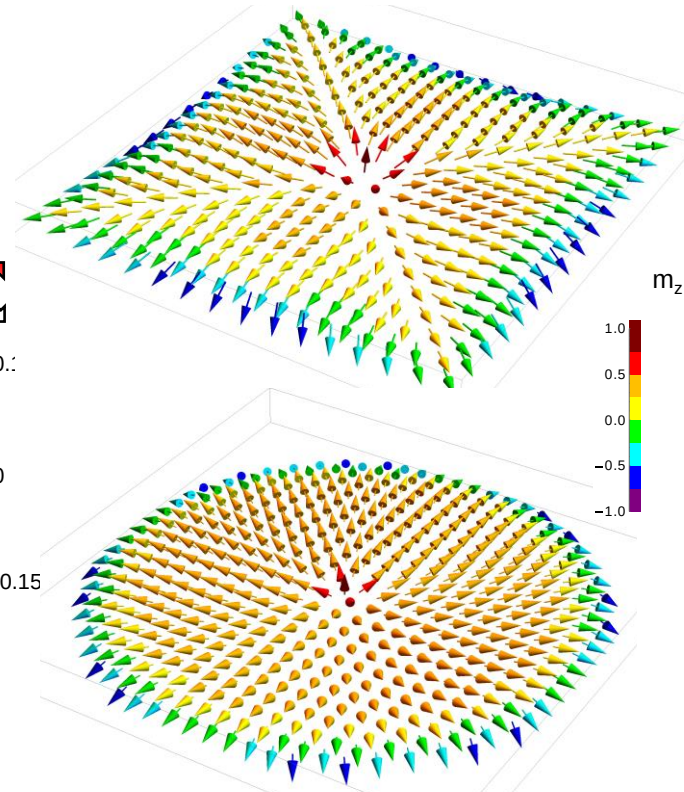
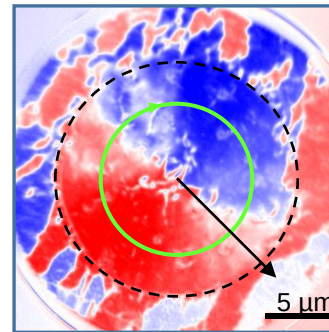
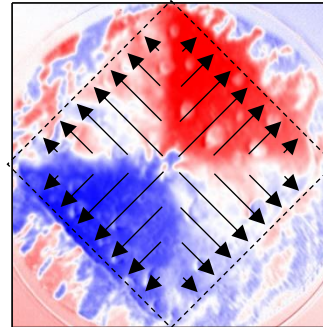
Encoded spin textures in FM / SC Hybrid Systems

A. Palau

YBCO-CSD film patterned with SC dots (20 μm x 20 μm) of different shapes, covered by a Soft FM



$T = 45\text{K}$ $B_{\text{max}} = +30\text{ mT}$



Versatile option to generate, modify and annihilate a large number of singular spin configurations for potential applications in logic or memory devices

Coated Conductors costumization for Integration in large scale devices

MINECO RTC (Gamesa /Siemens)

Ultra-high magnetic fields: Accelerators, Fusion, NMR, MRI...

← CERN Project
ALBA
IFAE
UPC

*Ultra-High field, 4.2K
Design stage, some prototypes*

**Energy generation:
Electrical Motors and
Generators (Wind Mills,
Airplanes, Ships,..)**

*High field, 40-60K
Much activity with prototypes*

**Energy efficiency
and transport :
cables, FCL,
transformers, ...**

*Low and medium field, 77K
Devices already in grid*

FASTGRID H2020
-NMBP (3 industries)

GAMESA project: MINECO-RTC-Wind Generator

X. Granados

“Design of a new generation of wind generator and auxiliary equipment for solar energy based on superconductors”

ICMAB
INSTITUT DE CIÈNCIA DE MATERIALS DE BARCELONA

EXCELENCIA
SEVERO
OCHOA

CSIC
CONSEJO SUPERIOR DE INVESTIGACIONES CIENTÍFICAS



Gamesa



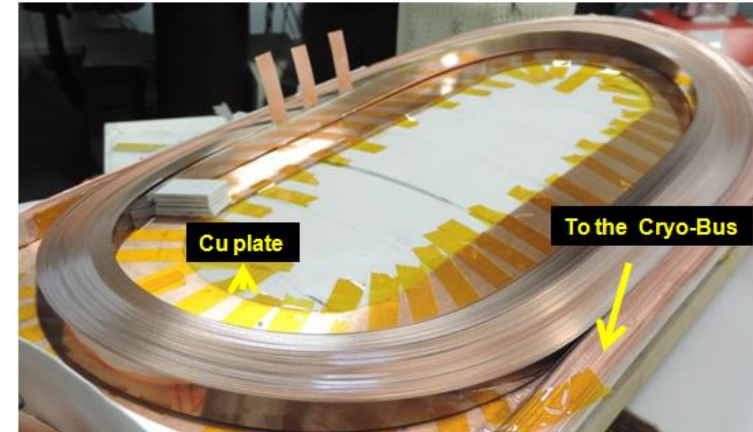
icma
Instituto de Ciencia
de Materiales de Aragón



Towards wind turbines generators based on HTS for next generation 10 MW



More efficiency
Less weight
Less volume



HTS coil

HTS stator for wind generator of 2 MW

Financing support phase 2 pending: HTS rotor (5 MW) (from 6.5 tm down to 2 tm)

FASTGRID H2020-NMBP

X. Obradors 12 partners (coord. CNRS, France)

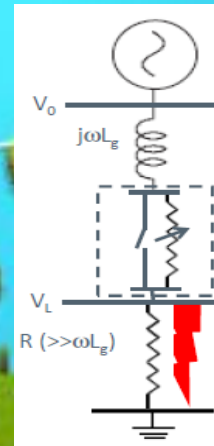
“Cost effective FCL using advanced superconducting tapes for future HVDC grids”



Smart DC SCFCL module (1 kA – 50 kV) for HVDC cables

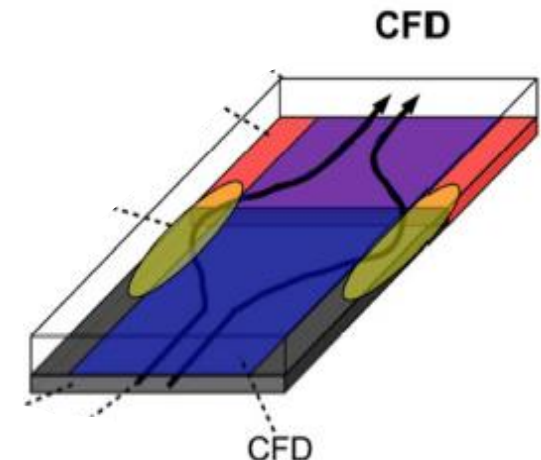
Novel architectures to modulate the superconducting transition

HVDC Supergrid concept



New materials concepts:

- New nanolayer at the surface to increase quench velocity (x10)
- New architecture based on sapphire to increase electric field (x20)

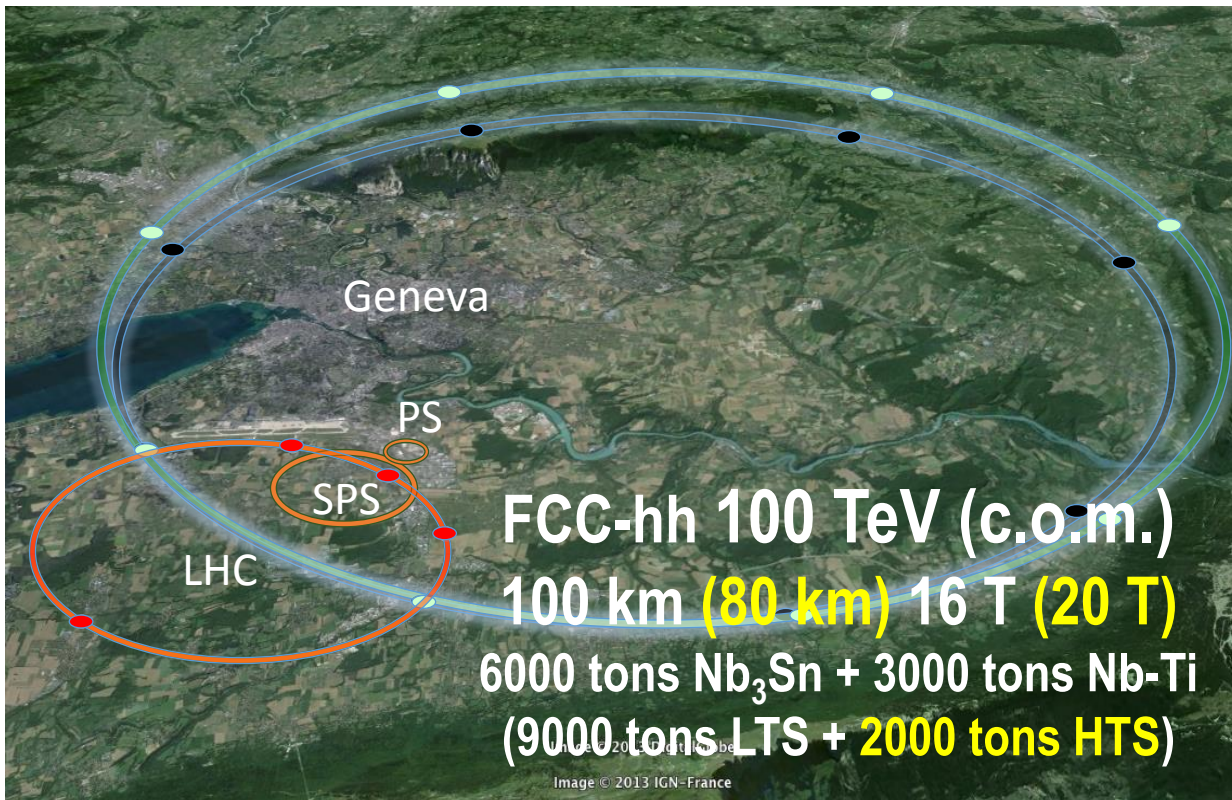


CERN collaborative project + FIP Proj. (SO)

J. Gutierrez, X. Granados, T. Puig

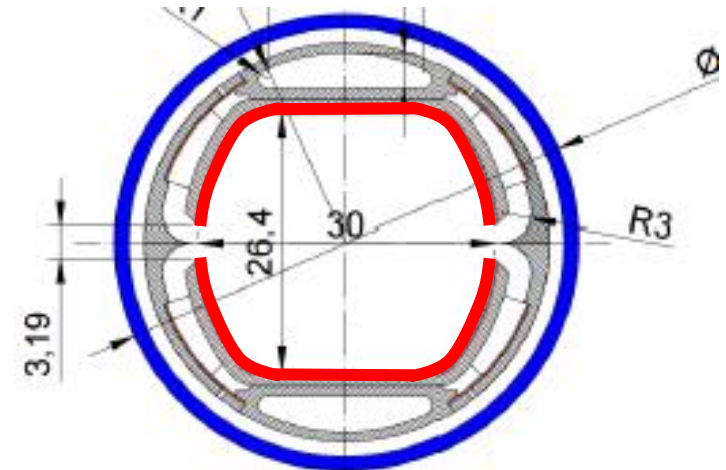
“HTS coated conductors for FCC beam screen”

Future Circular Collider (FCC)



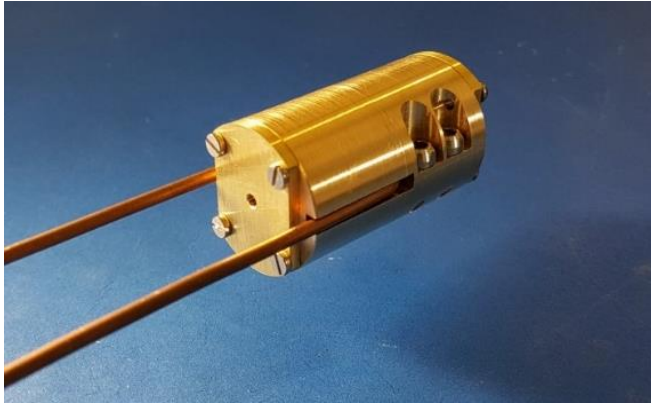
New HTS coated FCC chamber

Need of low surface impedance material for beam screen



CCs welded to stainless steel walls

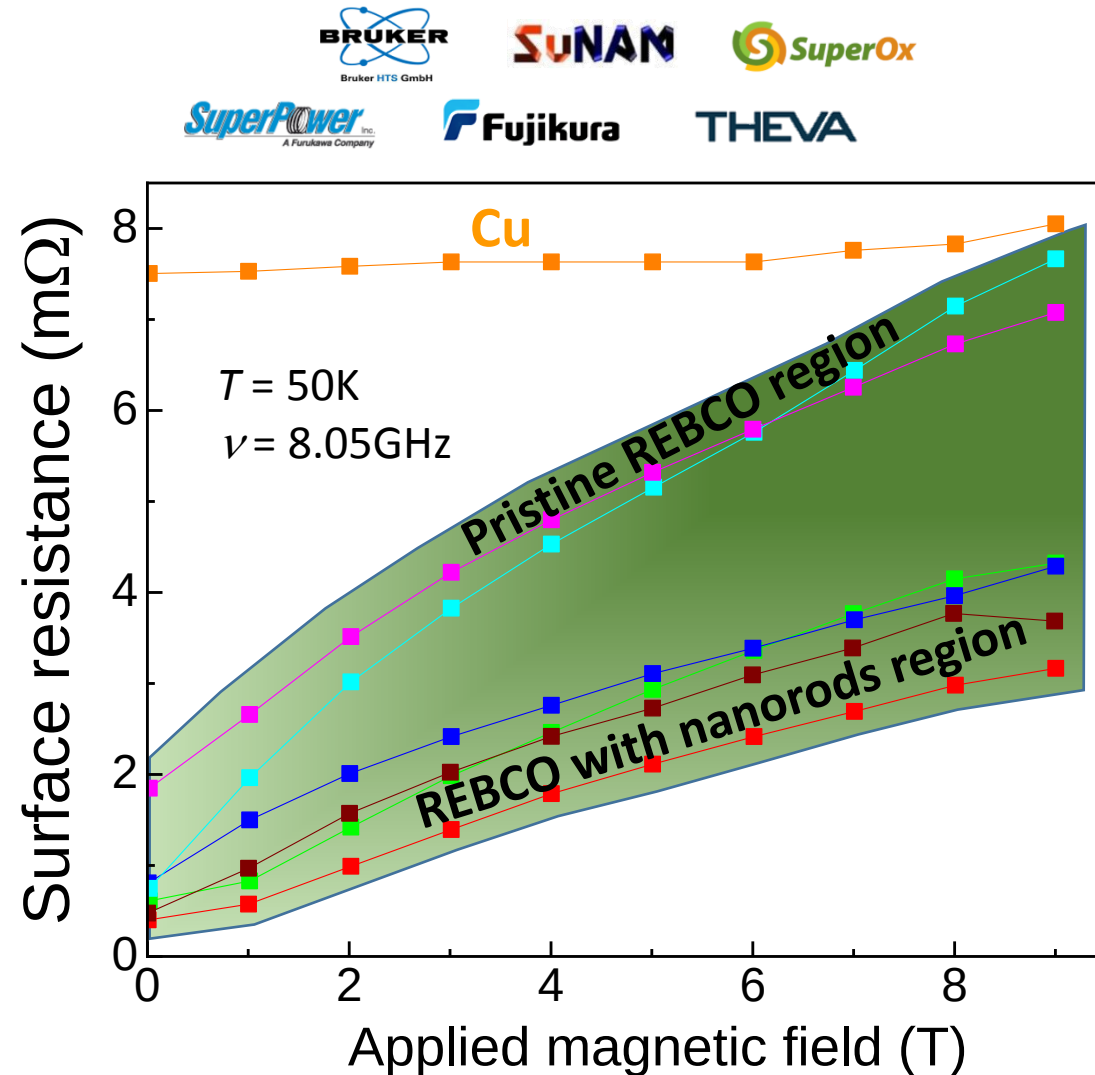
REBCO CCs show the potential to outperform Cu under the FCC working conditions



In house developed 8.05 GHz cavity resonator compatible with 25mm bore 9 T magnet at ICMAB

We have a 1 GHz resonator compatible with a 25 mm bore magnet in its final stage of design

REBCO CCs outperform Cu at 50K and up to 9T
 R_{sf} is microstructure dependent



CONCLUSIONS

The Superconductivity RL is strongly involved in:

- Materials development and scalability
- New superconducting functionalities
- Materials integration in devices

We want Superconducting materials and devices to penetrate the market

A group of physicist, chemists, materials science and engineers working together to make it possible

Cost/performance is the main challenge

We need Superconductivity dissemination to all levels



Welcome to the Superconducting Materials group